

Nonlinear Complementarity Problems

Chapter 8e: Applications and Theory

Based on Pathak's *Nonlinear Analysis and Fixed Point Theory*

July 14, 2026

Chapter Outline

① Introduction and Motivation

- Historical development and applications
- Problem formulations

② Complementarity Problem Formulations

- Explicit Complementarity Problem (E.C.P)
- Implicit Complementarity Problem (I.C.P)
- Simultaneous versions (S.E.C.P, S.I.C.P)

③ Theoretical Framework

- Cones and dual cones
- n -Lipschitz and n -strongly monotone mappings

④ Main Existence and Uniqueness Results

- Theorem 9.39 (Pathak et al.)
- Solution methods and convergence

⑤ Applications and Examples

- Optimization problems
- Engineering and economic applications

Historical Development of Complementarity Problems

- **Origins (1960s):** Complementarity problems emerged in the context of mathematical programming and optimization
- **Key Contributors:**
 - Cottle, Lemke: Linear complementarity problems
 - Karamardian: Nonlinear complementarity theory
 - Pang, Ferris: Variational inequality connections
- **Motivations:**
 - Optimization with constraints
 - Economic equilibrium problems
 - Contact mechanics and engineering
 - Structural analysis and elasticity
- **Modern Applications:**
 - Machine learning: Support vector machines
 - Game theory: Nash equilibrium
 - Variational inequalities
 - Friction and contact problems in mechanics

Wide Range of Applications

Complementarity problems have applications in:

Optimization Theory

- Constrained optimization
- Convex analysis
- Nonconvex optimization

Engineering

- Structural mechanics
- Elasticity theory
- Lubrication theory

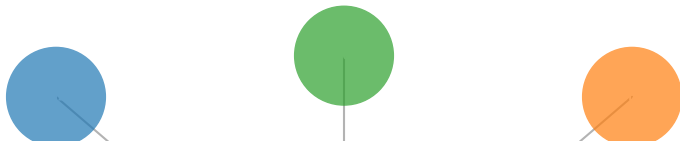
Economics and Game Theory

- Economic equilibrium
- Game theory
- Market equilibrium

Variational Problems

- Variational inequalities
- Differential equations
- Control theory

Applications of Complementarity Problems



Cones and Dual Cones

Definition

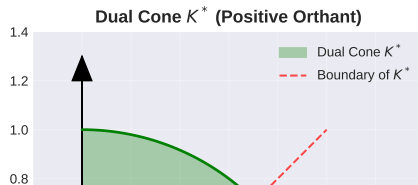
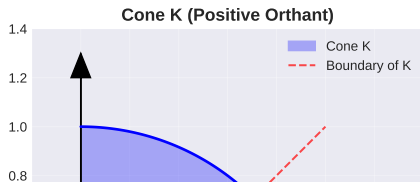
Let E be a locally convex space and $K \subseteq E$ a **closed convex cone**:

$$K^* = \{u \in E^* : \langle u, x \rangle \geq 0 \text{ for all } x \in K\}$$

is called the **dual cone** or **polar cone** of K .

Key Properties:

- K^* is always a closed convex cone
- If K is a closed convex cone, then $(K^*)^* = K$
- For the positive orthant: $(\mathbb{R}_+^n)^* = \mathbb{R}_+^n$



Complementarity Condition

Definition

Let K be a closed convex cone in a Hilbert space $\mathcal{H}$. Given mappings $f : K \rightarrow E^*$ and $g : K \rightarrow E$, we seek $x \in K$ such that:

$$\begin{aligned}f(x) &\in K^* \\g(x) &\in K \\ \langle g(x), f(x) \rangle &= 0\end{aligned}$$

Interpretation:

- $x \in K$: solution in the feasible region
- $f(x) \in K^*$: dual constraint satisfied
- $g(x) \in K$: primal constraint satisfied
- $\langle g(x), f(x) \rangle = 0$: complementarity (orthogonality condition)

Complementarity Condition: $\langle x, f(x) \rangle = 0$

1.4

- Example: $f(x) = e^{-x}$
- Region where $x \in K$
- Complementarity point

Explicit Complementarity Problem (E.C.P)

Definition (Explicit Complementarity Problem)

E.C.P: Find $x_0 \in K$ such that $f(x_0) \in K^*$ and

$$\langle x_0, f(x_0) \rangle = 0$$

Characteristics:

- Simplest form of complementarity
- Requires x_0 and $f(x_0)$ to be orthogonal
- Both must lie in their respective cones
- Widely studied in linear and nonlinear contexts

Example (Linear Complementarity):

$$z \geq 0, \quad w \geq 0, \quad z^T w = 0, \quad w = Mz + q$$

where M is a matrix and q is a vector.

Implicit Complementarity Problem (I.C.P)

Definition (Implicit Complementarity Problem)

I.C.P: Find $x_0 \in K$ such that $f(x_0) \in K^*$, $g(x_0) \in K$ and

$$\langle g(x_0), f(x_0) \rangle = 0$$

Characteristics:

- Generalizes E.C.P with auxiliary mapping g
- More flexible: orthogonality involves $g(x_0)$ instead of x_0 itself
- Applications to variational inequalities
- Encompasses many practical problems

Special Case: When $g = \text{identity}$, I.C.P reduces to E.C.P.

Simultaneous Complementarity Problems

Definition (Simultaneous Explicit Complementarity Problem)

S.E.C.P(f, g, K): Find $x_0 \in K$ such that $f(x_0) \in K^*$, $g(x_0) \in K^*$ and

$$\langle x_0, f^n(x_0) \rangle = 0 \quad \text{and} \quad \langle x_0, g^n(x_0) \rangle = 0$$

Definition (Simultaneous Implicit Complementarity Problem)

S.I.C.P(f_1, f_2, g, K): Find $x_0 \in K$ such that $f_1(x_0) \in K^*$, $f_2(x_0) \in K^*$, $g(x_0) \in K$ and

$$\langle g(x_0), f_1^n(x_0) \rangle = 0 \quad \text{and} \quad \langle g(x_0), f_2^n(x_0) \rangle = 0$$

Remarks:

- S.E.C.P contains E.C.P as a special case (when one function equals identity)
- S.I.C.P contains I.C.P as a special case (when one function equals identity)
- Both are fundamental in multifunction theory

Pairwise n-Lipschitz Mappings

Definition (Definition 9.10)

Let D be a subset of a Hilbert space $\mathcal{H}$, and consider mappings $f_1, f_2 : D \rightarrow \mathcal{H}$. We say f_1 and f_2 are **pairwise n -Lipschitz mappings with respect to g** if there exists $\beta > 0$ such that

$$\|f_1^n(x) - f_2^n(y)\| \leq \beta \cdot \|g(x) - g(y)\|$$

for all $x, y \in D$.

Key Aspects:

- Generalizes standard Lipschitz condition
- Involves iterate f^n (composition of f with itself n times)
- β is the Lipschitz constant
- Relates f to reference mapping g

Example: If $f(x) = 0.5 \cdot x$ and $g(x) = x$, then $f^n(x) = 0.5^n \cdot x$ is pairwise n -Lipschitz with $\beta = 0.5^n$.

Pairwise n-Strongly Monotone Mappings

Definition (Definition 9.11)

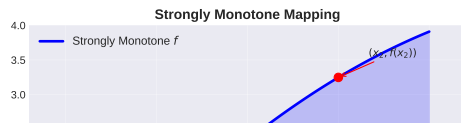
Let D be a subset of a Hilbert space $\mathcal{H}$, and consider mappings $f_1, f_2, g : D \rightarrow \mathcal{H}$. We say f_1 and f_2 are **pairwise n-strongly monotone mappings with respect to g** if there exists $a > 0$ such that

$$\langle f_1^n(x) - f_2^n(y), g(x) - g(y) \rangle \geq a \cdot \|g(x) - g(y)\|^2$$

for all $x, y \in D$.

Key Aspects:

- a is the strong monotonicity constant
- Implies strong contractivity in certain norms
- Guarantees uniqueness of fixed points
- Combined with Lipschitz condition gives existence and uniqueness



Theorem 9.39: Main Result (Pathak et al.)

Theorem (Theorem 9.39 - Pathak et al. 1996)

Let $\mathcal{H}$ be a Hilbert space and K a closed convex cone in $\mathcal{H}$. If, for a subset D of $\mathcal{H}$, the mappings $f_1, f_2, g : D \rightarrow \mathcal{H}$ satisfy the following:

- (1) f_1 and f_2 are pairwise n -strongly monotone with respect to g ,
- (2) f_1 and f_2 are pairwise n -Lipschitz with respect to g ,
- (3) There exists a real number $r > 0$ such that

$$r \cdot \beta^2 < 2a < \frac{1}{r} + r \cdot \beta^2$$

- (4) $K \subseteq g(D)$,

then the problem $S.I.C.P(f_1, f_2, g, K)$ is solvable. Moreover, if g is one to one, then the problem $S.I.C.P(f_1, f_2, g, K)$ has a unique solution in D .

Proof Strategy (Outline) I

Key Idea: Transform S.I.C.P into a fixed point problem.

Step 1: Complementarity to Fixed Point

- Define auxiliary mappings: $T_1^n(u) = P_K(u - r \cdot h_1^n(u))$ and $T_2^n(u) = P_K(u - r \cdot h_2^n(u))$
- where $h_i^n(u) = f_i^n(x)$ for $u = g^{-1}(x)$
- The complementarity problem is equivalent to finding a common fixed point

Step 2: Contractivity Analysis

- Show that $\|T_1^n(u) - T_2^n(u)\|^2 \leq \theta^2 \|u - v\|^2$
- where $\theta = (1 - 2r \cdot a + r^2 \cdot \beta^2)^{1/2} < 1$
- Use condition (3) to ensure $0 < \theta < 1$

Step 3: Apply Fixed Point Theorem

- By Lemma 9.4: existence and uniqueness of common fixed point in Hilbert space
- Complete metric space structure guarantees convergence

Proof Strategy (Outline) II

- Compose with g to get solution in original space

Step 4: Uniqueness via Injectivity

- If g is one-to-one: solution is unique in D
- Follows from uniqueness of common fixed point in K

Proof Strategy (Outline) III

Detailed Computation:

From the strong monotonicity condition (1):

$$\langle h_1^n(u) - h_2^n(v), u - v \rangle \geq \alpha \|u - v\|^2$$

From the Lipschitz condition (2):

$$\|h_1^n(u) - h_2^n(v)\| \leq \beta \|u - v\|$$

The projection operator satisfies:

$$\|T_1^n(u) - T_2^n(v)\|^2 = \|P_K(u - rh_1^n(u)) - P_K(v - rh_2^n(v))\|^2$$

Using projection properties:

$$\begin{aligned} &\leq \|u - v - r(h_1^n(u) - h_2^n(v))\|^2 \\ &= \|u - v\|^2 - 2r(u - v)^T(h_1^n(u) - h_2^n(v)) + r^2\|h_1^n(u) - h_2^n(v)\|^2 \end{aligned}$$

Proof Strategy (Outline) IV

Applying conditions (1) and (2):

$$\begin{aligned} &\leq \|u - v\|^2 - 2r \cdot a \|u - v\|^2 + r^2 \beta^2 \|u - v\|^2 \\ &= (1 - 2ra + r^2 \beta^2) \|u - v\|^2 \end{aligned}$$

Thus: $\|T_1^n(u) - T_2^n(u)\| \leq \theta \|u - v\|$ where $\theta = \sqrt{1 - 2ra + r^2 \beta^2} < 1$

Fixed Point Iteration Algorithm

Algorithm: Solving S.I.C.P(f_1, f_2, g, K)

- ➊ **Initialize:** Choose initial guess $x_0 \in K$, set $n = 0$, tolerance $\epsilon > 0$
- ➋ **Compute:**
 - $u_n = g(x_n)$
 - $h_1 = f_1^n(u_n), h_2 = f_2^n(u_n)$
 - $u_{n+1} = P_K(u_n - r \cdot h_1) \cap P_K(u_n - r \cdot h_2)$
 - $x_{n+1} = g^{-1}(u_{n+1})$
- ➌ **Check:** If $\|x_{n+1} - x_n\| < \epsilon$, stop; else set $n := n + 1$ and go to step 2
- ➍ **Return:** $x^* \approx x_n$ as solution

Convergence Properties:

- Linear convergence: $\|x_n - x^*\| \leq C\theta^n$ for some $C > 0$
- Rate depends on $\theta = \sqrt{1 - 2ra + r^2\beta^2}$
- Choice of r optimized using condition (3)

Numerical Example: Simple 2D Complementarity Problem

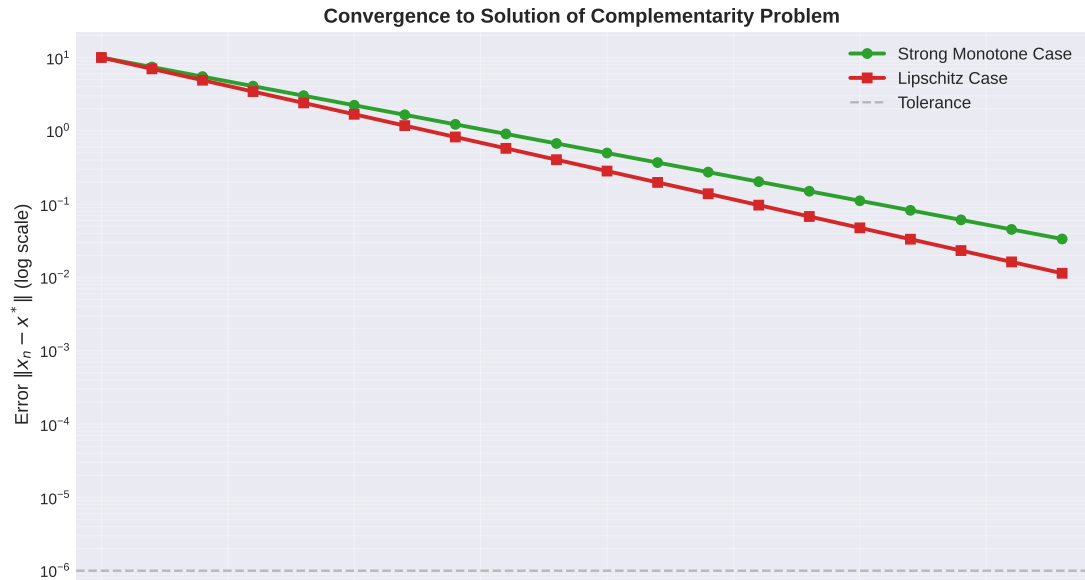
Problem: Solve E.C.P with $K = \mathbb{R}_+^2$ (positive orthant)

Consider: $f(x) = \begin{pmatrix} 2x_1 + x_2 - 1 \\ x_1 + 2x_2 - 1 \end{pmatrix}$

Complementarity condition: $x \in K, f(x) \in K^*, \langle x, f(x) \rangle = 0$

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 # Define objective function
5 def objective(x):
6     return (2*x[0] + x[1] - 1)**2 + (x[0] + 2*x[1] - 1)**2
7
8 # Complementarity violation
9 def complementarity_violation(x):
10     f = np.array([2*x[0] + x[1] - 1, x[0] + 2*x[1] - 1])
11     return np.dot(x, f)
12
13 # Constraints: x >= 0, f(x) >= 0
14 constraints = [
15     {'type': 'ineq', 'fun': lambda x: x[0]},
16     {'type': 'ineq', 'fun': lambda x: x[1]},
17     {'type': 'ineq', 'fun': lambda x: 2*x[0] + x[1] - 1},
18     {'type': 'ineq', 'fun': lambda x: x[0] + 2*x[1] - 1}
19 ]
```

Convergence Behavior



Application 1: Variational Inequalities

Definition

A **variational inequality problem** (VIP) is to find $x \in K$ such that

$$\langle f(x), y - x \rangle \geq 0 \quad \forall y \in K$$

Connection to Complementarity:

- VIP is equivalent to: find $x \in K$, $f(x) \in K^*$ with $\langle f(x), x \rangle = \langle f(x), y \rangle$ for all $y \in K$
- For convex K , this is equivalent to E.C.P under certain conditions
- Many algorithms for VIP convert to complementarity form

Example (Projection Problem): Minimize $\|x - a\|^2$ subject to $x \in K$

Solution: $x^* = P_K(a)$ where P_K is the projection onto K

Complementarity form: Find $x \in K$ such that $\langle x - a, y - x \rangle \geq 0$ for all $y \in K$

Application 2: Economic Equilibrium

Market Equilibrium Problem:

- Determine equilibrium prices $p \in K = \mathbb{R}_+^n$
- Excess demand function $f(p)$ representing supply-demand imbalance
- Equilibrium: either market clears ($f(p) = 0$) or good is free ($p_i = 0 \Rightarrow f_i(p) \leq 0$)

Complementarity Formulation:

$$p \in K, \quad f(p) \in K^*, \quad \langle p, f(p) \rangle = 0$$

Economic Interpretation:

- If $p_i > 0$: good i is traded, so $f_i(p) = 0$ (market clears)
- If $f_i(p) > 0$: excess demand, so $p_i = 0$ (good is free)
- Never simultaneously: complementarity enforces mutual exclusivity

Applications: Trade equilibrium, labor markets, auction design

Application 3: Mechanics and Contact Problems

Contact Problem (Unilateral Constraint):

- Two elastic bodies in contact
- Normal force $\lambda \geq 0$ at contact
- Gap $g(u) \geq 0$ (bodies separated or touching)
- Complementarity: $g(u) \cdot \lambda = 0$

Mathematical Formulation: Find displacement u and normal force λ such that:

$$u \in \text{admissible set}$$

$$\lambda \geq 0 \quad (\text{non-negative force})$$

$$g(u) \geq 0 \quad (\text{gap constraint})$$

$$g(u) \cdot \lambda = 0 \quad (\text{complementarity})$$

Physical Interpretation:

- If bodies are separated ($g > 0$): no contact force ($\lambda = 0$)
- If in contact ($g = 0$): can have positive force ($\lambda \geq 0$)
- Cannot have both gap and force simultaneously

Applications: Friction, adhesion, crack propagation, robot manipulation

Application 4: Optimization with Complementarity Constraints

Bilevel Optimization Problem:

Minimize $\Phi(x, y)$ subject to:

$$\text{(upper level)} \quad x \in X$$

$$\text{(lower level)} \quad y \in K, \quad f(x, y) \in K^*, \quad \langle y, f(x, y) \rangle = 0$$

Characteristics:

- Optimization problem with complementarity constraint
- Constraint is a complementarity condition (nonsmooth)
- Applications in game theory, inverse problems, data fitting

Example (Inverse Problem):

- Recover parameters from partial observations
- Solution must satisfy optimality conditions (complementarity)
- Often solved using penalty methods or augmented Lagrangian

Challenge: Constraint set is typically non-convex; requires specialized algorithms

Special Cases of the Complementarity Framework

- **Case 1:** $n = 1$ and $g = \text{identity}$
 - S.I.C.P reduces to I.C.P (implicit complementarity)
- **Case 2:** $n = 1$, $g = \text{identity}$, $f_1 = f_2$
 - Reduces to I.C.P with single function
- **Case 3:** $g(x_0) = x_0 - m(x_0)$ (**point-to-point mapping**)
 - Problem S.I.C.P(f_1, f_2, g, K) relates to finding $x_0 \in K$ such that

$$\langle x_0 - m(x_0), f_1(x_0) \rangle = 0 \text{ and } \langle x_0 - m(x_0), f_2(x_0) \rangle = 0$$

- **Case 4:** $g = \text{identity mapping}$
 - Problem reduces to standard simultaneous complementarity
 - f_1 and f_2 are pairwise strongly monotone/Lipschitz

Strongly Nonlinear Quasi-Complementarity

Remark

The strongly nonlinear quasi-complementarity problem discussed by Siddiqi and Ansari is a particular case of the simultaneous complementarity framework.

Definition (Strongly Nonlinear Quasi-Complementarity):

Find $x \in K$ such that there exist $f(x) \in K^*$ and $g(x) \in K$ satisfying:

$$\langle g(x), f^2(x) \rangle = 0$$

where $f^2(x) = f(f(x))$ is the second iterate.

Relationship to Our Framework:

- This is a special case where we iterate mappings
- Can be viewed as a fixed point of composed operators
- Existence proven using monotonicity and Lipschitz properties
- Standard fixed point theory applies

Remarks on Problem Variants

Remark (Remark 9.10)

For $f_1 = f_2$ and $n = 1$, problem S.E.C.P(f_1, f_2, K) contains E.C.P as a particular case.

Remark (Remark 9.11)

For $f_1 = f_2$ and $n = 1$, problem S.I.C.P(f_1, f_2, g, K) contains I.C.P as a particular case.

Remark (Remark 9.12)

If we take $n = 1$ and $g(x_0) = x_0 - m(x_0)$ where m is a point-to-point mapping from E into itself, then S.I.C.P(f_1, f_2, g, K) reduces to the problem of finding $x_0 \in K$ such that $f_1(x_0) \in K^*$, $f_2(x_0) \in K^*$ and

$$\langle x_0 - m(x_0), f_1(x_0) \rangle = 0 \text{ and } \langle x_0 - m(x_0), f_2(x_0) \rangle = 0$$

Remark (Remark 9.13)

If we take $n = 1$ and g the identity mapping on K , then problem S.I.C.P(f_1, f_2, g, K) reduces to the problem of finding $x_0 \in K$ such that $f_1(x_0) \in K^*$, $f_2(x_0) \in K^*$ and

Exercises I

- ❶ **Exercise 9.1:** Use Banach's fixed point theorem to show that the system

$$\begin{aligned}x_1 &= \frac{1}{3}x_1 - \frac{1}{4}x_2 + \frac{1}{4}x_3 - 1 \\x_2 &= -\frac{1}{5}x_1 + \frac{1}{6}x_2 + \frac{1}{5}x_3 + 2 \\x_3 &= \frac{1}{5}x_1 - \frac{1}{3}x_2 + \frac{1}{4}x_3 - 2\end{aligned}$$

has a unique solution.

- ❷ **Exercise 9.2:** Show that the initial value problem

$$\begin{cases} \frac{du}{dt} = u^2, & t \geq 0 \\ u(0) = 0 \end{cases}$$

has more than one solution.

- 8 **Exercise 9.3:** Show that the solutions of the initial value problem

$$\frac{dx}{dt} = |x|^{1/2}, \quad x(0) = 0$$

are $x_1(t) = 0$ and $x_2(t) = \frac{t^2}{4}$. Does this contradict Picard's theorem? Find other solutions.

Exercises III

- 4 **Exercise 9.4:** Show that an initial value problem given by

$$\left\{ \begin{array}{l} \frac{d^2 x}{dt^2} = f(t, x), \quad x(t_0) = x_0, \quad x'(t_0) = x_1 \end{array} \right.$$

involving a second-order differential equation can be transformed into a Volterra integral equation.

- 5 **Exercise 9.5:** Consider the Volterra integral equation

$$f(t) = g(t) + \int_0^t K(t, s)f(s) ds, \quad a < t < b$$

where K is continuous on $[a, b] \times [a, b]$. Show that the equation has a unique solution $f \in C[a, b]$ for each fixed $g \in C[a, b]$.

6 **Exercise 9.6:** Consider the Fredholm integral equation

$$f(t) = g(t) + \lambda \int_a^b K(t, x) f(x) dx$$

where $g \in L_\infty[0, 1]$ and $K \in L_2([0, 1] \times [0, 1])$. Prove that if $g = 0$ implies $f = 0$, then there exists a unique solution for any $g \in L_\infty[0, 1]$.

🔗 **Exercise 9.7:** Consider the nonlinear integral equation

$$v(t) = x(t) - \lambda \int_a^b K(t, s, x(s)) ds$$

where v and K are continuous on $[a, b]$ and $[a, b] \times [a, b] \times \mathbb{R}$, respectively, and K satisfies on $[a, b] \times [a, b]$ a Lipschitz condition of the form

$$|K(t, s, u_1) - K(t, s, u_2)| \leq c|u_1 - u_2|$$

Then, show that

$$|\lambda| < \frac{1}{c(b-a)}$$

Exercises VI

- 8 **Exercise 9.8:** Approximate numerical solutions of $f(x) = 0$. To find approximate numerical solutions of $f(x) = 0$, we may convert the equation in the form $x = g(x)$, choose an initial value x_0 and compute

$$x_n = g(x_{n-1}), \quad n = 1, 2, \dots$$

Suppose that g is continuously differentiable on some interval $J = [x_0 - r, x_0 + r]$ and satisfies $|g'(x)| \leq c < 1$ on J and $|g(x_0) - x_0| < (1 - c)r$. Show that the iterative sequence $\{x_n\}$ converges to a solution and the error estimates are:

$$\begin{aligned} (i) \quad & |x - x_n| < c^n r \\ (ii) \quad & |x - x_n| \leq \frac{c}{1 - c} |x_n - x_{n-1}| \end{aligned}$$

- 9 **Exercise 9.9:** Newton's method. Let f be real-valued and twice continuously differentiable on an interval $[a, b]$. Let $\bar{x}$ be a simple zero of f in $[a, b]$. Show that Newton's method defined by

$$x_{n+1} = g(x_n), \quad g(x_n) = x_n - \frac{f(x_n)}{f'(x_n)}$$

is a contraction in some neighborhood of $\bar{x}$ so that the iterative sequence converges to $\bar{x}$ for any x_0 sufficiently close to $\bar{x}$.

Summary and Key Takeaways

Core Concepts:

- **Complementarity problems** provide a unifying framework for optimization, variational inequalities, and equilibrium problems
- **Four formulations:** E.C.P, I.C.P, S.E.C.P, S.I.C.P with increasing generality
- **Key properties:** pairwise n -Lipschitz and n -strongly monotone mappings ensure existence and uniqueness

Main Result (Theorem 9.39):

- Provides sufficient conditions for solvability
- Uses fixed point theory in Hilbert spaces
- Guarantees unique solution when g is injective
- Convergence rate depends on contractivity factor θ

Applications:

- Variational inequalities and optimization
- Economic equilibrium and market models
- Mechanical contact and friction problems
- Bilevel optimization and inverse problems

Further Reading and References

Core References:

- Pathak, H. K. (1996). *An Introduction to Nonlinear Analysis and Fixed Point Theory*. Singapore: Springer Nature.
- Cottle, R. W., Pang, J. S., & Stone, R. E. (2009). *The Linear Complementarity Problem*. SIAM.

Classical References:

- Karamardian, S. (1976). Complementarity problems over cones with monotone and pseudomonotone maps. *Journal of Optimization Theory and Applications*, 18(4), 445-454.
- Pang, J. S. (1991). A posteriori error bounds for the linear complementarity problem. *Mathematics of Operations Research*, 12(3), 368-392.

Advanced Topics:

- Variational inequalities and monotone operators
- Fixed point theory in metric spaces
- Multivalued mappings and inclusions
- Proximal algorithms and operator splitting